Vision and Scope Document

for

Student Management System (SMS)

Version 1.0 approved

Prepared by Team 1

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Revision History

Name	Date	Reason For Changes	Version
Team 1	3/25/26	Initial draft	1.0 draft 1
Team 1	3/25/26	Baseline approved	1.0 approved

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1. Business Requirements

1.1. Background

Currently, student management at the school is done manually through paper documents and fragmented Excel spreadsheets. Academic staff spend an average of 4 hours per day cross-checking course registration data and entering grades. Students frequently report difficulties understanding their study plan and registering for elective courses due to the outdated or overloaded system.

1.2. Business Opportunity

Developing an online management system (SMS) will help automate administrative processes, minimize data errors, and enhance academic experience for students. The system will provide a centralized platform for managing records, grades, and course registrations, helping the school optimize its teaching resources.

1.3. Business Objectives

BO-1: Reduce the processing time for administrative procedures related to course registration by 50% within 6 months of implementation.

BO-2: Increase the productivity of administrative staff by 20% through report automation.

BO-3: Achieve 100% digitization and secure storage of student grade data and records.

1.4. Success Metrics

SM-1: At least 90% of students used the system to register for courses online in the first semester of its implementation.

SM-2: Student satisfaction with training services increased from 3.0 to 4.5 on a 5.0 scale within 12 months.

1.5. Vision Statement

For students and administrators struggling with manual management processes, the Student Management System (SMS) is an online educational management platform that intelligently and securely connects academic records, course registration, and learning outcomes management. Unlike current fragmented processes, SMS provides real-time data, enabling students to proactively manage their learning paths and helping the school make more accurate decisions.

1.6. Business Risks

RI-1: Faculty or staff may be hesitant to switch from paper-based to digital processes (Probability = 0.4, Impact = 5).

RI-2: The system may experience network infrastructure issues during peak course registration periods (Probability = 0.3, Impact = 8).

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1.7. Business Assumptions and Dependencies

AS-1: All students and staff have access to a computer or mobile device with internet connectivity.

DE-1: The SMS depends on the availability of the University's central database for initial student data migration.

2. Scope and Limitations

2.1. Major Features

FE-1: Course registration (register/add/drop) with prerequisite, credit-limit, timetable-conflict, and seat-capacity checks.

FE-2: Input midterm/final grades (Lecturer grade entry and submission, with validation).

FE-3: View timetable (Student/Lecturer personal timetable viewing).

FE-4: Open course sections (Academic Admin creates/publishes sections, assigns lecturers, sets timetable/room and capacity).

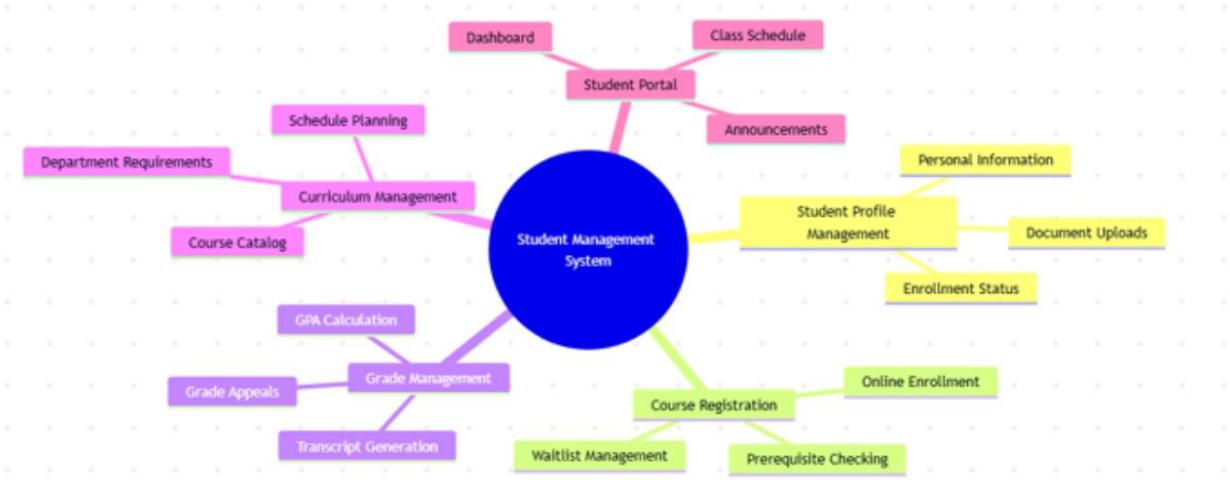

Figure 1. Partial feature tree for the Student Management System.

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2.2. Scope of Initial and Subsequent Releases

Feature	Release 1	Release 2	Release 3
FE-1, Course Registration	Core registration (add/drop, validations)	Enhancements (TBD)	Enhancements (TBD)
FE-2, Input Midterm/Final Grades	Grade entry and submission	Enhancements (TBD)	Enhancements (TBD)
FE-3, View Timetable	View personal timetable	Enhancements (TBD)	Enhancements (TBD)
FE-4, Open Course Sections	Create/publish sections; assign lecturers; set timetable/room & capacity	Enhancements (TBD)	Enhancements (TBD)

2.3. Limitations and Exclusions

LI-1: Release 1.0 will not include online tuition fee payment (planned for Release 2.0).

LI-2: The system is exclusively for full-time undergraduate students; short-term certificate programs are excluded.

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3. Business Context

3.1. Stakeholder Profiles

Stakeholder	Major Value	Attitudes	Major Interests	Constraints
University Management	Improve training management efficiency; ensure accurate reporting data; save operational costs.	Strongly support digitalization to catch up with ed-tech trends.	Data security; stable system operation and intuitive reporting.	Limited project budget; the system must comply with current training regulations.
Academic Admin	Automate authorization, class creation, and record management processes; minimize manual paperwork.	Supportive but concerned about data entry and migration from the old system.	Easy user management; ability to quickly handle course registration issues.	Require system training; time pressure during digital transformation.
Lecturers	Save time managing class rosters and entering grades; reduce data errors.	Positive reception but desire a minimalist, easy-to-use interface.	Simplicity of use; reliable access; quick grade entry; auditability.	Accuracy of grade data; quick grading interface, draft saving support.
Students	View timetables, grades, and register for courses anytime, anywhere; high transparency.	Very enthusiastic but will be highly strict about page loading speed during course registration.	The system doesn't crash during registration; friendly interface on both desktop and mobile.	Must have internet-connected devices; need pre-issued accounts from the Admin.

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3.2. Project Priorities

Dimension	Constraint	Driver	Degree of Freedom
Features	Must complete 4 core Use Cases: Register for Courses, View Timetable, Input Midterm/Final Grades, Open Course Sections for Release 1.0.		Can skip secondary features like: Online tuition payment, automated academic warnings.
Quality	No fatal bugs in the Course Registration flow and password security.	Friendly, visually appealing interface (React.js).	Minor UI/UX display bugs on mobile devices are acceptable for the demo version.
Schedule	Must submit SRS and Vision & Scope documents before 11:00 PM, Tuesday, March 31, 2026.	Complete Frontend and Backend deployment before the project defense date.	
Cost			Student projects should use free tools and servers (e.g., host frontend on Vercel, use free databases).
Staff		The team has only 3 members (FE, BE, Documentation).	Flexible cross-support (the documentation person can help the dev team with testing once done).

3.3. Deployment Considerations

The Student Management System (SMS) will be deployed using a React.js frontend hosted by CDN and a Java Spring Boot backend. The system will utilize PostgreSQL (via Supabase) with Prisma ORM for data management, alongside Redis for distributed locking to prevent race conditions during peak course registration. Advanced integrations, including WebSockets for real-time notifications, must be fully operational for the initial release. Initial database configuration and mock data migration must be completed prior to launch. Finally, a training video of no more than five minutes in length should be developed to guide users on the web-based system.

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